

A systematic review of solar irradiance forecasting across time horizons using physical, satellite, and AI-based methods

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ABSTRACT

Solar energy is a key renewable resource, yet synchronizing its production with grid demand remains challenging due to irradiance variability. This review systematically examines the evolution of solar irradiance forecasting methods, from physical and numerical weather prediction (NWP) models to modern data-driven and hybrid AI-based approaches. It integrates analyses of All-Sky Imager (ASI) and satellite datasets and categorizes forecasting techniques by temporal horizons from intra-hour to multi-day forecasts. The paper uniquely contributes a comparative taxonomy linking forecast horizon, input data type, and model architecture to accuracy outcomes. The findings highlight the growing benefits of physics-informed deep learning for improving operational solar forecasts.

1. Introduction

With the continuous growth in population, the importance of energy supply has escalated significantly. It is anticipated that in the coming years, energy supply will become a paramount concern for countries. Despite the high emissions and finite nature of fossil fuels, their accessibility has led to a substantial reliance on them. Therefore, it is crucial to identify and develop suitable alternatives, particularly renewable energy sources.

Solar energy is one of the renewable resources that are easily accessible in most parts of the world [1]. Despite the increasing adoption of this energy source and its relatively recent development compared to other renewable sources, the challenges associated with its full utilization remain unresolved.

After the Kyoto Protocol, a coherent policy toward renewable energies has been put in place globally in order to slow down the pace of global warming [2]. A large number of developing and developed countries are investing in solar energy in order to comply with the Paris Agreement and relieve the extreme effects of global warming. A major downside of renewable energies, particularly solar energy, is related to uncertainties about the power generation capacity of solar systems in the upcoming days [3].

Over the last ten years, the capacity for electricity production from

photovoltaic sources has surged by >900 gigawatts, with investments in this sector rising by approximately 19 %. In 2021 alone, solar energy production capacity increased by 180 gigawatts. Key factors contributing to this progress include the reduction in storage battery costs and the expansion of local battery storage systems, particularly in solar farms and decentralized photovoltaic production [4].

The primary challenges in the solar energy sector currently involve the unpredictability of incoming radiation and the amount of solar energy generated over specific periods. Efforts must be made to mitigate these uncertainties [5].

Based on the weather conditions of a specific geographic area, predictions can be made regarding incoming solar radiation. However, due to potential changes and inaccuracies in the input weather data, there are inherent errors in these predictions. Accurate forecasting of incoming radiation is crucial to reduce costs and minimize the impact of forecast errors on ratepayers, grid operators, wind and solar farm maintenance crews and energy traders. These errors can lead to price increases, project shutdowns, or financial losses. Therefore, it is essential to strive for the most accurate predictions of wind and solar energy [6].

Given the inherently intermittent nature of solar energy, accurately forecasting its generation trends is of paramount importance. Reliable power generation forecasts are widely recognized by policymakers as a key instrument for mitigating availability challenges and ensuring grid stability [7,8]. To achieve optimal photovoltaic system utilization,

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Nomenclature			
AI	Artificial Intelligence	CNN	Convolutional Neural Networks
ANN	Artificial Neural Network	ML	Machine Learning
AR / ARMA / ARIMA	Autoregressive / Autoregressive Moving Average / Autoregressive Integrated Moving Average	MLP	Multilayer Perceptron
ASI	All-Sky Imager	MOS	Model Output Statistics
CSP	Concentrated Solar Power	SVM/SVR	Support Vector Machine / Support Vector Regression
CSI	Clear Sky Index	NWP	Numerical Weather Prediction
DNI	Direct Normal Irradiance	PCA	Principal Component Analysis
ECMWF	European Centre for Medium-Range Weather Forecasts	PM	Persistence Model
GFS	Global Forecast System	PV	Photovoltaic
GHI	Global Horizontal Irradiance	RF	Random Forest
MAE	Mean Absolute Error	RMSE	Root Mean Square Error
MAPE	Mean Absolute Percentage Error	rRMSE	Relative Root Mean Square Error
MBE	Mean Bias Error	nRMSE	normalized Root Mean Square Error
nMBE	normalized Mean Bias Error	CRMSE	Central Root Mean Square Error
		TSI	Total Sky Imager
		WRF	Weather Research and Forecasting Model
		R ²	Coefficient of determination

highly accurate forecasts are required. These forecasts minimize network fluctuations across multiple temporal horizons (long-term, medium-term, and short-term) [9].

For clarity of scope, this review adopts the following classification of forecast horizons:

- Very-short / Intra-hour (up to 60 min): Primarily used for now-casting, this range supports real-time voltage regulation, rapid energy-storage response, and short-term power smoothing.
- Short / Intra-day (1–6 h): Suitable for local dispatch optimization, near-term generation forecasting, and coordination with distributed storage resources.
- Day-ahead (6–48 h): Critical for unit commitment, market bidding, and resource allocation within operational planning cycles.
- Multi-day / Seasonal (beyond 48 h up to several weeks): Pertinent to maintenance scheduling, long-term planning, and strategic generation management [10].

This temporal differentiation is crucial because forecasting models, data inputs, preprocessing procedures, and evaluation metrics may vary substantially depending on the horizon considered.

In this systematic review, we explore the evolution of solar irradiance forecasting techniques aimed at improving solar energy generation prediction. The reviewed methods are organized into three overarching categories:

- Physical and Numerical Weather Prediction (NWP)–based approaches
- Classical statistical and machine-learning techniques
- Modern deep-learning and hybrid frameworks

We compare these approaches across different forecast horizons, highlighting their respective strengths, limitations, and domains of applicability. Furthermore, emerging paradigms such as physics-informed learning frameworks, probabilistic and ensemble forecasting, and data-fusion methodologies integrating satellite and all-sky imager (ASI) observations are discussed to illuminate future research directions.

Accordingly, the paper is structured as follows: Section 2 presents an overview of solar radiation measurement techniques and forecasting methods. Section 3 outlines the key environmental and meteorological factors influencing solar irradiance. Section 4 details the various forecasting models, followed by an analytical comparison within Section 4. Finally, Section 5 concludes the study and highlights perspectives for future work.

2. Measurement and forecasting

Despite the challenges in forecasting solar radiation, solar energy remains the most widely utilized renewable energy source, largely due to its seamless integration with the electricity supply network [11].

The success of any solar project hinges on the availability of accurate global solar radiation data. Unfortunately, due to the high cost of measuring equipment and technical limitations in calibration, only a limited number of weather stations worldwide record such data. In response to this challenge, researchers have explored alternative data generation methods. This exploration could inspire the development of robust forecasting techniques that integrate the strengths of existing models.

The operation and maintenance of photovoltaic (PV) plants are crucial for professionals in the solar industry. Accurate prediction of PV output power is essential for effective electricity planning, distribution, optimal management, quality assurance, and grid stability. However, designing precise models for PV output power prediction is challenging due to the inherent uncertainties caused by variations in solar radiation. Forecasting methods can be broadly categorized into four groups: physical models (such as numerical weather prediction models), statistical techniques (such as AR, ARMA, ARIMA), artificial intelligence (AI) methods (including machine learning (ML) and deep learning (DL)), and hybrid approaches that combine two or more methods.

When considering different time scales, intra-hour forecasts (up to 1 hour) enhance grid quality and stability, while intraday forecasts (up to 6 h) are crucial for optimal integration. One-day forecasts are mainly used for unit commitment planning and dispatching capacity, whereas one-week forecasts are applicable in trading, management, and maintenance [12]. Beyond the aforementioned scenarios, over time, additional methods such as those based on sensor networks are employed, and their performance, errors, and advantages are thoroughly examined [13]. In numerous countries, extensive research has been conducted to identify optimal solar radiation prediction models tailored to the specific climatic conditions of each region [14,15].

3. Factors affecting incoming solar radiation

Multiple factors influence the amount of radiant energy in each geographical area. These factors have been studied extensively in various articles, considering their individual effects and potential interactions. The key factors are stated in the following.

3.1. Particulate matter

Particulate matter in the atmosphere are one of the most important factors affecting the solar radiation of the earth's surface. In an study, the impact of aerosols on the precision of solar radiation forecasting was examined using a high-resolution, one-day-ahead operational numerical weather prediction (NWP) model. To isolate the effect of aerosols, the research focuses on clear sky periods. Severe events attributed to aerosols, such as dust or fog, were identified during these periods. Thus, accurate aerosol forecasting plays a crucial role in enhancing solar radiation prediction accuracy [16].

Short-term (2–3-day) solar radiation forecasts play a crucial role in solar energy industries, particularly for yield forecasting when integrating highly variable solar energy into the power grid. In regions and time periods focused on solar energy, cloud-free conditions prevail. During these times, aerosols become the main atmospheric factor affecting direct and global irradiance at the Earth's surface. For example, in a five-month period in Europe, the accuracy of aerosol optical depth at 550 nm (AOD550) predictions was assessed using particle forecasts from the CTM European Air Pollution Dispersion (EURAD) model [17].

Following an assessment of day- and hour-ahead solar irradiance prediction applications within the energy industry, aerosol optical depth (AOD) forecasts at 550 nm derived from a chemistry transport model were found to surpass ground-based measurements. Notably, sporadic desert dust storm events in the central Mediterranean region were identified as the primary source of significant inaccuracies in the model predictions. These AOD predictions, along with other remote sensing data (like Earth albedo and ozone) and numerical weather prediction parameters (such as water vapor and clouds), were used as inputs for a radiation forecasting system [18].

3.2. Cloud cover

Another important factor that have impact on the performance of a PV system, is the cloud covering of the sky. A 15-minute advance notice before cloud cover is adequate for activating backup systems and minimizing the reliance on electrical storage during clear-sky conditions. As a test case, it is essential to assess satellite-based cloud and radiation data, conduct atmospheric modeling of aerosol loading, and perform numerical weather prediction for clouds and radiation using deterministic modeling and ensemble techniques. These efforts align with the requirements for integrating solar power into the grid [19]. For example, Stratocumulus clouds (Sc) significantly influence surface solar resources along the California coast, particularly where rooftop photovoltaic systems are prevalent. Their extensive spatial coverage, optical thickness, and reflectivity contribute to this variability. However, current numerical weather prediction models lack the capability to predict both the presence and thickness of Sc. Consequently, validation efforts focus on a specific summer month in southern California, when Sc clouds are most common [20].

In an investigation of the impact of cloud intensity on the predicted global horizontal irradiance (GHI), a novel approach was introduced. This approach utilized satellite imagery and ground measurements as input data. Specifically, it relied on cloud tracking and transmission using an optical flow algorithm. Subsequently, at the chosen location (San Diego, California), the estimated cloud transport along semi-steady streamlines was calculated. Based on this method, radiation forecasts were obtained. Fig. 1 illustrates the data processing flowchart for the solar forecast system, which relies on the satellite radiation model [21]. Table 1 presents the calculated error results from this research. For comparison, SolarAnywhere results were included.

One challenge in predicting solar energy is the variability in solar penetration due to cloud cover. The Total Sky Imager (TSI) is a short-term irradiance forecasting method with a 5-minute forecast horizon for low-altitude clouds. Jin Xu et al. [22] suggested integrating NWP into

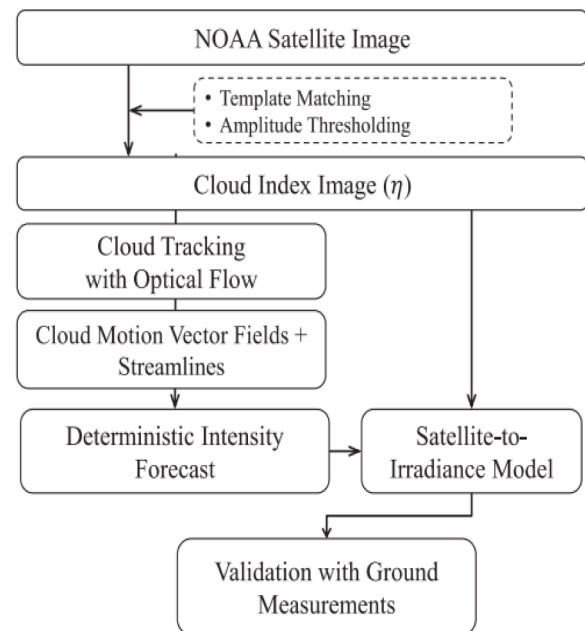


Fig. 1. Schematic of data processing for the satellite-to-irradiance model based solar forecasting system [21].

TSI to extend the forecast horizon to 15 min. This approach aimed to detect multi-layer clouds and track identified cloud segments. To achieve this, three basic models with different feature combinations were used: the Persistence Model (PM), Linear Regressions (LRs), and Support Vector Machines (SVRs). The performance of each model, shown in Fig. 2 based on their Root Mean Square Error (RMSE), indicated that the SVR model performed the best, resulting in an effective irradiance forecasting model for 1–15 min predictions.

Taking into account several variables, including the clear sky index (Kc), solar zenith angle (SZA), and the cloud pattern in solar radiation prediction, it is observed that the absolute error increases as SZA and Kc decrease. Notably, the RMSE is notably affected by the mean and standard deviation of Kc [23].

Numerical climate prediction models frequently underestimate cloud cover, which results in an overestimation of GHI. By integrating cloud water path (CWP) data from geostationary satellites using an Ensemble Kalman Filter (EnKF), short-term GHI forecasts from the Weather Research and Forecasting (WRF) model have improved in mid-latitude case studies [24].

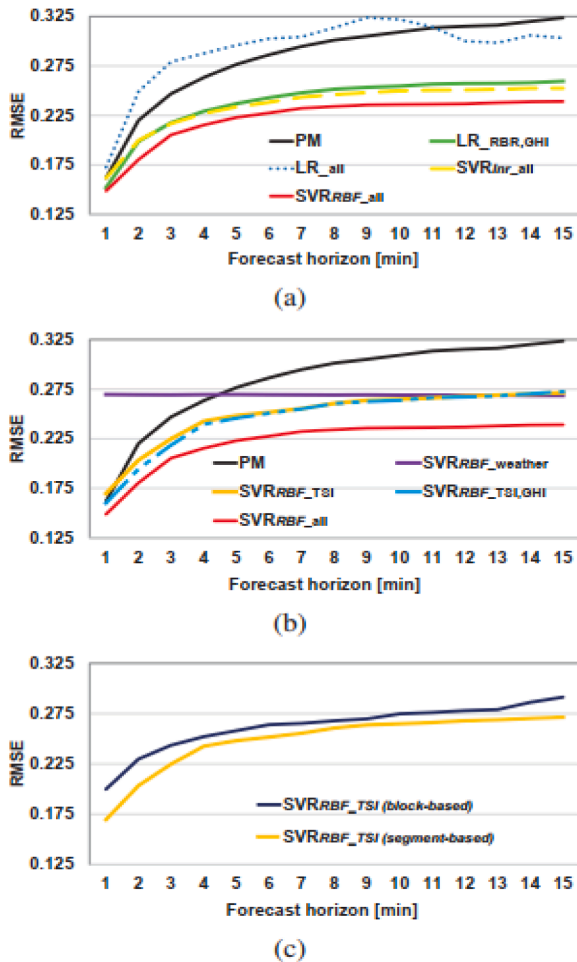
In recent years, there has been ongoing development of machine learning models aimed at analyzing surface radiation variability in relation to cloud type and cloud fraction. These models employ various machine learning approaches, including multivariate linear regression, k-nearest neighbors (kNN), random forests (RF), gradient boosting machines (GBM), and multilayer perceptrons (MLP or ANN). The input features for these models encompass a measure of solar variability and the ratio of observed all-sky irradiance to estimated clear-sky irradiance, referred to as effective transfer (ET). Fractional sky cover (FSC) acts as a continuous predictor variable, whereas cloud type (CT) is considered a categorical variable [25].

Despite the challenges posed by cloud cover in solar radiation forecasting, various solutions are being explored to mitigate this error. One effective tool for short-term radiation forecasting is the total sky imager, which currently provides data at 5-minute intervals primarily due to low-altitude clouds. Efforts to extend its predictive horizon include integrating weather data from NWP at intervals ranging from 15 min to several days into a short-term radiation prediction model. This integration has resulted in an average improvement of 21 % compared to the base model [26]. Models are employed to reconstruct historical total

Table 1

Comparison between results from the satellite-to-irradiance models and available data from the SolarAnywhere [22].

San Diego	Mean measured (W/m ²)	Mean modeled (W/m ²)	MBE (W/m ²)	rMBE (%)	MAE (W/m ²)	rMAE (%)	RMSE (W/m ²)	rRMSE (%)	Xcor (-)
NOAA GHI Model 1	493.49	495.73	-2.25	-0.46	71.72	14.53	124.53	25.24	0.934
NOAA GHI Model 2	493.49	516.78	-23.29	-4.72	61.72	12.51	107.41	21.77	0.95
SUNY GHI	385.31	455.16	-69.86	-18.13	106.40	27.61	213.14	55.32	0.825
SUNY GHI (outliers removed)	415.95	439.89	-23.95	-5.76	64.71	15.56	130.52	31.38	0.932

**Fig. 2.** Comparison of different models (a), feature sets (b), and cloud tracking pipelines (c) for 1–15 min forecast [22].

solar radiation (TSI) when direct space-based measurements are unavailable [27].

3.3. Humidity

The sky transparency index, derived from relative humidity profiles across three atmospheric sub-layers, significantly influences global solar radiation. It serves as a link between solar radiation under clear sky conditions and cloud cover. For instance, research on the Tibetan Plateau indicated that convective activity in the region introduces uncertainty in radiation estimates due to the localized and transient nature of these systems [28].

3.4. Wind velocity

Numerous studies have investigated the impact of wind intensity on incoming solar radiation and the joint assessment of these two

parameters. For instance, Federica Davo et al. [29] employed Principal Component Analysis (PCA) along with two post-processing techniques to predict wind power in Sicily and assess solar radiation at specific measurement locations.

3.5. Ozone air quality

The precision of the model's solar radiation forecasts is greatly influenced by the aerosol optical depth. When this depth surpasses 0.1, the model's errors can be as high as 100 W/m². Conversely, for lower aerosol optical depths, the models' use of climatological attenuation results in solar irradiance estimates that are very close [30].

3.6. Earth

Rugged terrain introduces variability in the incoming solar radiation field, yet its impact remains inadequately accounted for in most numerical weather prediction models. Specifically, for a horizontal surface, reduced radiation has been demonstrated to decrease the root mean square error of the WRF model by approximately 20 % (equivalent to about 25 W/m²) during winter and autumn, and by 60 W/m² during spring and summer. Meanwhile, for a tilted surface, downscaling to different spatial resolutions yielded the best results at a resolution of 9 arcseconds, with a root-mean-square error of 45 % (equivalent to 57 W/m²) and an average bias error close to zero [31].

In numerous studies, researchers have explored the combined impact of various factors. For instance, Zhian Sun et al. [32] developed a rapid estimation scheme for assessing instantaneous direct solar radiation (DSI) at the Earth's surface. Additionally, the influence of dust varies based on soil quality and this can clearly have impact on the solar radiation [33].

4. Models for predicting solar radiation

4.1. Physical models

Developing models for accurately measuring solar radiation involves incorporating various meteorological parameters (such as latitude, longitude, and altitude) and climatic factors (including sunshine duration, humidity, transparency index, seasonal variations, temperature, cloudiness, wind speed, atmospheric pressure, radiation diffusion, global radiation, extraterrestrial radiation, and evaporation) [34]. Physical methods include three main approaches: sky imagery [35], NWP [36], and satellite-imaging [21]. The accuracy of physical models is significantly influenced by the interactions between the dynamic movement of solar radiation and the physical state. Given that photovoltaics and concentrated solar power (CSP) are increasingly contributing to global energy resources, there is a substantial demand for reliable solar energy forecasting. These forecasts are primarily generated by physical or statistical models, both of which depend on precise solar radiation predictions. For short- to medium-term forecast horizons (hours to days), radiation forecasts are predominantly provided by NWP models. However, despite recent advancements aimed at enhancing irradiance prediction in current NWP models, their quality remains suboptimal and is a major source of uncertainty in PV power prediction.

A promising approach to improve NWP solar radiation forecasting is the multimodel method [37].

NWP models begin with a description of the atmospheric conditions at a specific time. They use equations that describe horizontal and vertical momentum, hydrostatic continuity, and energy conservation, capturing atmospheric variability over fixed time steps. There are two categories of NWP models based on the spatial extent of their modeling domain. The first category includes global models, which cover the entire globe, such as the European Centre for Medium-Range Weather Forecasts (ECMWF) and the Global Forecast System (GFS). The second category includes regional or mesoscale models, which cover specific regions. Examples of these are the Weather Research and Forecasting (WRF) model, the Mesoscale Atmospheric Simulation System (MASS) model, the Advanced Multiscale Regional Prediction System (ARPS), the North American Model (NAM), and the Fifth-generation Pennsylvania State University-National Center for Atmospheric Research Mesoscale (MM5) model [38].

To broaden the application of this model, various solutions and methods have been explored for its implementation. Each method aims to reduce forecasting time from days to hours and minutes, while enhancing accuracy across different locations.

In 2014, short- and medium-term solar radiation forecasts were introduced to evaluate the accuracy of numerical weather forecasting models. Numerical models, such as the air pollution model (TAPM), can estimate GHI using standard meteorological fields, though they lack real-time cloud and aerosol data. To assess the model's performance, the GHI estimated by TAPM was compared with satellite-derived GHI at selected sites. The results showed that the 45 km resolution had the lowest RMSE, indicating that the model struggles to accurately capture changes on scales smaller than 45 km [39].

To address these shortcomings, Mathiesen et al. [40] introduced a high-resolution cloud absorption model at the University of California. Using the Weather Research and Forecasting (WRF) model, they implemented an operational 1.3-km grid solar forecast for San Diego, California, optimized to accurately simulate local meteorological conditions, especially fog and marine layers during summer. They developed a direct cloud assimilation system (WRF-CLDDA) to predict intra-hourly variability and produce accurate cloud field initializations. By utilizing satellite imagery and ground weather station reports, WRF-CLDDA statistically adjusts cloud hydrometeors (water content and cloud water vapor) in the initial conditions.

Although numerical weather forecasting models are generally more accurate than image-based or statistical models, they tend to over-predict radiation, leading to meteorological errors. This over-prediction, particularly in regions with dynamic cloud systems, reduces forecast accuracy [41]. While these models are significant in their applications, they generally operate on a scale much larger than what is effective for network integration, particularly for accurate radiation prediction.

In a research in this context, to enhance variability forecasts derived from the 10 km coarse forecast, various methodologies, including analog downscaling a pattern matching algorithm utilizing a historical forecast library to identify groups similar to current forecasts are being explored. This process necessitates WRF version 3.6.2 to produce a series of 10 km + 2 km nested simulations. Radiation data, accurate to ± 2 %, was sourced from the SURFRAD network. As indicated in Table 2, only the WRF-simulated GHI aligns with the observed radiation [42].

Also, Wong et al. [43] conducted an investigation into global average daily radiation by examining solar radiation models for predicting global average daily and hourly irradiance, beam irradiance, and diffuse irradiance. Seven models were considered, utilizing the Angstrom-Prescott equation to predict global average daily radiation based on sunshine hours in Hong Kong. Additionally, two parametric models were investigated to estimate the monthly average global hourly radiation and forecast the hourly radiation and a comparative analysis was conducted between model predictions and measured data.

Table 2

Analog weights and correlation coefficients [42].

Variable	w-GA	Correlation	Descriptive Name
GHI	0.37	0.89	Global Horizontal Irradiance
qC	0.35	0.01	Columnar Cloud Water Mixing Ratio
qR	0.19	0.00	Columnar Rain Water Mixing Ratio
qI	0.06	0.00	Columnar Ice Mixing Ratio
Prec	0.02	0.05	Surface Precipitation
qV _{2m}	0.01	0.22	2-m Water Vapor Mixing Ratio
Qv	0.00	0.13	Columnar Water Vapor Mixing Ratio
T _{2m}	0.00	0.40	2-m Temperature

Subsequently, to enhance the accuracy of time step examinations, Junior et al. [44], introduced a method for predicting solar radiation one hour in advance, utilizing numerical weather forecasts and recent measured data. Ultimately, irrespective of the spatial scale, the proposed method's predictions significantly outperformed the reference predictions, demonstrating consistent performance. The skill scores ranged from 0.67 to 0.73 for local scales and 0.97 for regional scales. To examine solar radiation forecasts at 10-minute intervals, Ian Sosa-Tinoco [45] developed a method for producing global solar radiation assessments with high spatial (9 km) and temporal (10 min) resolution. This method provided numerical weather forecasts based on a model. The simulation outputs were used to create radiation maps and the grid cells were compared with corresponding observations to evaluate the model's accuracy, achieving a correlation higher than 0.88 with the observational data.

Although NWP models can forecast solar radiation variables, they often exhibit significant errors, particularly in direct normal irradiance (DNI), which is highly influenced by aerosol and cloud type and concentration. One approach to enhancing 10-minute forecasts for DNI, as proposed by Pereira et al. [46], involves two artificial neural network (ANN) models. The first model utilizes weather and aerosol variables predicted at a specific moment, while the second model incorporates a period of improved DNI forecasts prior to the forecast moment. This research leverages data from the ECMWF and the Copernicus Atmospheric Monitoring Service. To enhance forecasting conditions and reduce the time required from one hour to one minute, Far Nanandez et al. [47] developed a method to describe the movements and high-frequency dynamics of DNI through the non-dimensionalization of solar radiation data. The results, based on two-year surveys across various climate points, indicate that the Kolmogorov-Smirnov integral parameter (KSI %) and the normalized root mean square values are below 23 % and 22 % per month, respectively, showing minimal discrepancies between modeling and measurement.

In many high-altitude regions of the Earth, the typical coarse resolution of models and the variable physical characteristics of the environment pose challenges for the NWP method. For instance, on Réunion Island in the southwest Indian Ocean basin, these issues are evident. To address this, Jordi Badosa et al. [48] investigated the influence of large-scale atmospheric parameters on local solar radiation and its daily variability. To enhance forecast accuracy on the island, they introduced an Empirical-Statistical Downscaling (ESD) method, which relies on identifying similar past conditions to predict the previous day's solar radiation. Using 12 years of ECMWF reanalysis data and measurements from 21 stations on the island, their results demonstrated that the ESD method outperforms climatology-based radiation predictions, improving daily RMSE by 4–11 %. The set of NWP methods can also be utilized to estimate solar radiation and wind power density, either in combination or separately [49].

4.1.1. Overview of physical models

Numerical Weather Prediction is one of the most effective forecasting methods, though selecting the appropriate model from the diverse range available remains a significant challenge. Consequently, numerous comparisons have been conducted to evaluate different NWP models

based on criteria such as RMSE and mean absolute error (MAE).

Verbois, et al., assessed 108 different models using a multi-criteria hybrid approach, Fourier analysis, and classification of daily radiation profiles based on various WRF mesoscale radiation parameters. This comprehensive evaluation enhanced the understanding of model distinctions and accuracy, highlighting the importance of multiple factors beyond model errors. The study concluded that while WRF forecasts were inadequate for hourly radiation resolution, they were more suitable for predicting daily radiation profiles [50].

In a separate study, Perez evaluated two NWP models the ECMWF global model and the WRF model based on the medium-scale GFS across three geographical regions: the United States, Canada, and Europe. The findings from all three studies indicate that the GFS-based WRF model forecasts are less accurate compared to global forecast-based approaches like the ECMWF, and also, a simple averaging of model outputs yields better results than using individual models [51].

The primary challenge in solar radiation forecasting lies in elucidating the nonlinear relationship between observed and predicted radiation. To address this, a transformation function is required to effectively communicate this relationship. According to Kakumoto et al. [52], the predicted NWP and observed radiation data were utilized to construct this function using multidimensional kernel density estimation (KDE). The probabilistic prediction of this joint function, derived through Bayes' law, offered superior statistical performance and reliability compared to other methods. This proposed method, termed Joint Probability Condition (CoJD), employs the mean value in place of scattered data and considers the clear sky index as a direct substitute for radiation in measuring solar radiation. Typically, CoJD involves two main phases: the construction phase and the forecasting phase. During the construction phase, a Joint Probability Density Function (PDF) is created between the predicted Clear Sky Index (CSI) and the observed CSI using a two-dimensional KDE method. In this study, the forecasting phase applies the Joint PDF to EnNWP to generate a forecast PDF for CSI. Fig. 3 illustrates this approach. The result of the forecast method studied at every site is shown in Fig. 4. For all the sites CoJD was better than EnReg an PP. In market trade simulation, CoJD also performed better than the other forecast method.

Verbois et al. [53] utilized NWP to study solar irradiance in tropical region. Fig. 5 depicts a network of 25 meteorological stations across Singapore, managed by the Solar Energy Research Institute of Singapore (SERIS). The evaluation was performed with a 1-hour time resolution; for instance, the hourly GHI for 9 am was calculated by averaging multiple values from 8:31 am to 9:30 am, with any missing minute-data being disregarded. In the next step, they used a high resolution

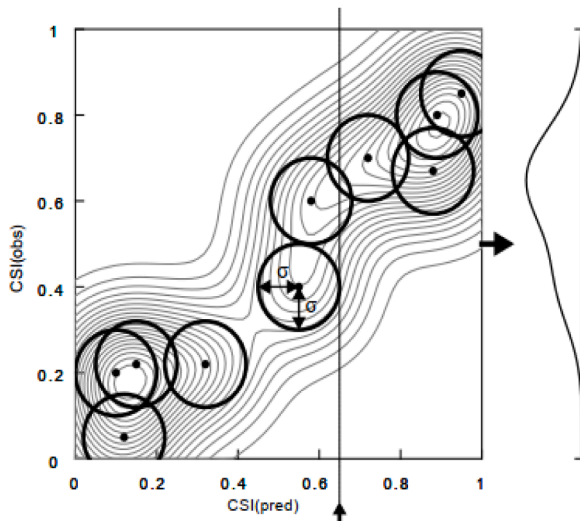


Fig. 3. Schematic illustration of CoJD [52].

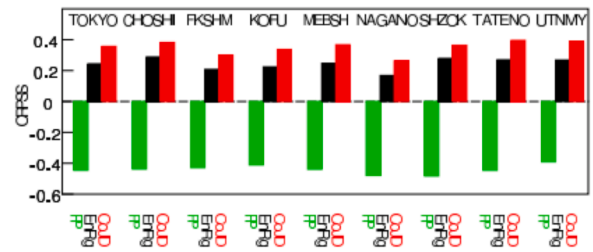


Fig. 4. Comparison of forecast methods [52].

implementation of WRF solar consisted of four involute domains with horizontal spatial resolution of 27, 9, 3 and 35 vertical levels. Lasso model output statics (Lasso), Analog Ensemble (AnEn) and Quantile Gradient Boosting (QGB) were used as post – processing models. Lasso and QGB both used all of the variables but AnEn only use a few of them. Briefly, WRF had the least accuracy, QGB had the lowest MAE which means QGB had more accuracy but comparing to Lasso model had too much outliers. After QGB and Lasso, AnEn had the lowest bias.

Over the years, advanced methods for forecasting solar radiation have been developed, utilizing outputs from NWP models in conjunction with radiative transfer operators like libRadtran. In an study, WRF-Solar configuration of the Weather Research and Forecasting (WRF) model was employed, which is specifically tailored for solar energy applications. Parameters forecasted by WRF, such as vertical profiles of cloud liquid water (CLW), ice water content (IWC), cloud fraction, relative humidity, and temperature, are utilized as inputs for libRadtran to compute offline solar radiation. At each forecast interval, Gaussian Mixture Model (GMM) clustering was used to classify all horizontal networks. The classification results showed that the networks are categorized into air masses or disturbances with comparable meteorological properties. Satellite observation data served as references. The mean square error (MSE) is broken down into the mean square of the error and the MSE of variables centered at zero. The square root of the latter is known as the central root mean square error (CRMSE). Forecast data was analyzed over a 2-day period. The mean error did not show significant change until the second day, while CRMSE remained consistent only on the first day. This framework clarified the structure of surface solar radiation forecast errors, aiding validation within the NWP model integration domain for any evolving air mass or disturbance [54].

In addition to generating electricity through large and small photovoltaic power plants, accurately calculating the received solar energy in urban areas is crucial. This urban-specific model, referred to as WRF-solar (uWRF-solar), demonstrates a strong predictive capability for GHI, with an R^2 value of 0.93 under clear sky conditions, 0.61 under cloudy sky conditions, and 0.39 under overcast conditions [55].

4.1.2. NWP model accuracy

Over the years, extensive efforts have been undertaken to identify the factors influencing the accuracy of NWP models. For instance, Perez et al. [56] evaluated seven GHI prediction models that depended directly or indirectly on NWP. To validate these models, ground measurements, including hourly integrated GHI recorded over a one-year period, were analyzed from seven SURFRAD network stations. The seven NWP-based GHI models evaluated in this study include the Global Multiscale Environmental Model (GEM), the ECMWF model, two versions of the Weather Research and Forecasting (WRF) model, the MASS model, the Advanced Multiscale Regional Prediction System (ARPS) model, and a model based on cloud cover forecasts from the US National Forecast Database (NDFD). These models provide forecasts up to 48 h, except for the ECMWF model, which forecasts up to three days, and the NDFD model, which forecasts up to seven days. The models were compared with ground measurements using RMSE, MAE, MBE, and the Kolmogorov-Smirnov integral (KSI). Models based on the European or Canadian Global Climate Simulation (GEM & ECMWF) demonstrated

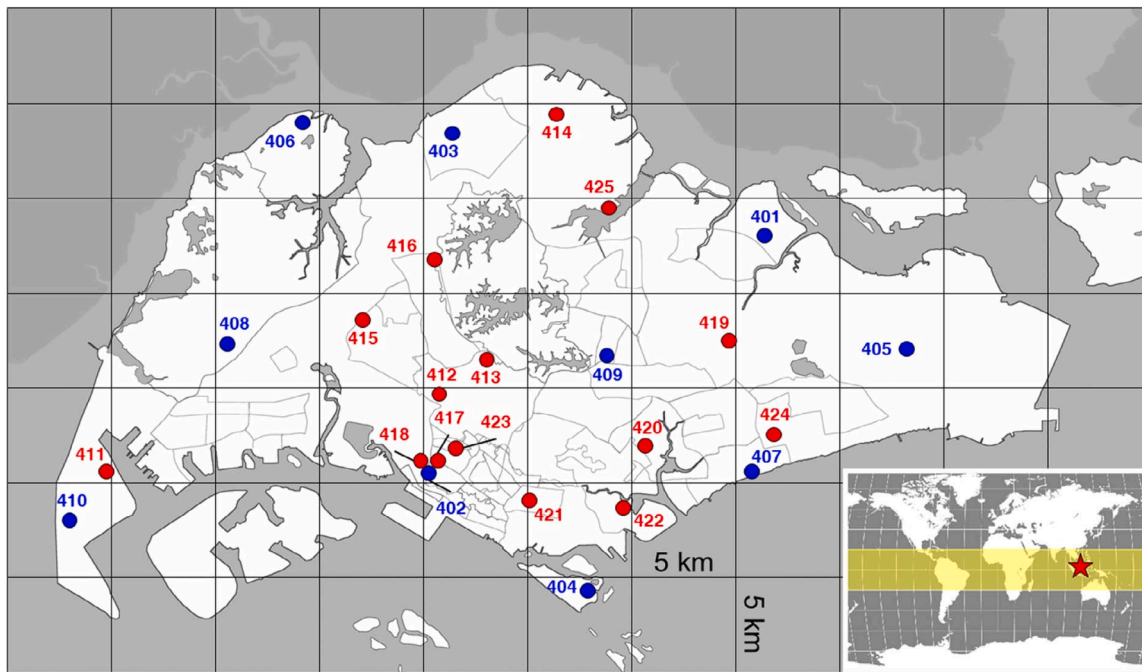


Fig. 5. SERIS network of meteorological stations operational since May 2013 [53].

superior RMSE results. Combining GEM and ECMWF models yielded better outcomes compared to using them individually. GFS-based models were tested with and without MOS postprocessing, which corrects local errors. Tests incorporating MOS showed improved performance, while non-MOS experiments highlighted the limitations of GFS-based model radiation algorithms.

The necessity to accurately predict solar radiation is driven by the initial and ongoing costs associated with all stages of solar energy utilization. Minimizing forecasting errors ensures that costs remain close to actual values. Sun [57] proposed an effective plan for predicting solar radiation under various weather conditions, relying on components of both clear and cloudy skies. Input variables were assessed through climate conditions and satellite observations within the NWP model, which serves as an offline plan for GHI calculation using satellite data. The results confirmed that this study can reduce net solar radiation error from 800 W/m^2 to $<50 \text{ W/m}^2$. Additionally, results indicated that the average half-hour relative error of GHI is reduced to $<7 \%$ under different sky conditions.

Upon examining the factors contributing to the inaccuracy of NWP models, it was identified that the type of cloud cover in the study area is a significant factor. For instance, marine stratus clouds are a primary source of inaccuracy in NWP models for coastal California. Proper parameterization, along with accurate initial datasets, significantly enhances the accuracy of simulated cloud fields and surface solar radiation [58]. Accurate solar radiation forecasting significantly contributes to precise operating cost estimations. However, achieving higher accuracy entails additional costs. In a research, to balance the optimal error margin with the costs of calculations and implementation, a three-hour forecasting period was considered. This interval may introduce errors in radiation forecasts, and simulations with lower resolution at high frequency may also experience inaccuracies due to humidity and velocity variations caused by atmospheric flow and clouds [59].

To mitigate the impact of cloud cover on NWP models, improvements are necessary. Consequently, many researchers have contributed to enhancing forecast accuracy by developing various approaches. For example, to reduce errors caused by cloud variability, researchers suggested that a model can be developed by first adopting a seasonal approach and then analyzing cloud change patterns that affect radiation

in the target area. Using a seasonal and climate-dependent approach (MOS), this model can reduce the relative root mean square error (rRMSE) of GHI by 6 % compared to the original model, demonstrating the effectiveness of an intelligent model [60].

PV modules are influenced by the spectral distribution of solar radiation. While Numerical Weather Prediction models compute radiation across multiple spectral bands, the standard outputs for PV forecasting typically provide only bandwidth values. This raises concerns about how the absence of detailed spectral data impacts PV power estimation. To investigate this, a physical PV model was created to determine the spectral distribution of radiation and the panels' spectral response. This model was tested using data collected at the Site Instrumental de Recherche par Télédétection Atmosphérique (SIRTA) in Palaiseau, France. The average relative difference between the simulated and measured PV power for a 250 W monocrystalline silicon module is -0.9% , with an average bias of -2.0 W . This model was employed to explore how solar zenith angle and cloud cover affect PV module performance. The results revealed that in cloudy conditions, PV performance can improve by up to 5 % through spectral filtering of near-infrared radiation, and by up to 18 % when only beneficial radiation is considered in performance calculations. Overall, global broadband horizontal irradiance is inadequate for precise PV power modeling, highlighting the importance of spectrally and angularly refined irradiances. This underscores the importance of PV models using more precise irradiance inputs, which can be easily obtained by utilizing the internal variables of NWP models [61]. By integrating Data Assimilation (DA) with Numerical Weather Prediction model observations, the issues related to incomplete initial atmospheric information for predictions can be mitigated in both static (3DVAR) and time-evolving (4DVAR) modes. This method was evaluated for its effectiveness in forecasting hourly conditions for the next day, and when applied to the WRF model, it demonstrated minimal error in time evolution under all conditions except clear skies. For instance, the initial 4DVAR forecasts, after processing, exhibited superior performance with a relative RMSE of 37 % over the Singapore study area [62].

In another study, to enhance the model's accuracy, a method involving the integration of three data sets, i.e., NWP model data, satellite survey data and ground-collected information, was employed. The

implementation of this method required three steps including Classification and Clustering (utilizing the Gaussian Mixture Model (GMM) to classify and cluster horizontal grid cells within the study area), CDF Analysis (Applying the Expectation-Maximization (EM) algorithm to examine the cumulative distribution functions (CDFs) of the transparency index (CI) derived from satellite observations and NWP data for predicting hourly radiation) and Correction Function Determination (determining the corresponding correction functions based on the inferred CDFs for each region clustered in step 1). Finally, corrections were applied to the CI of the NWP forecast data for days (d) and ($d + 1$) (Fig. 6). Mean Bias Error (MBE), RMSE, and Mean Absolute Error (MAE) were utilized to assess forecasting accuracy. Consequently, the algorithm demonstrated a superior correction effect in spring and summer compared to winter, by accounting for seasonal behavior [63].

Although NWP models provide a robust physical basis for solar irradiance prediction, their accuracy is limited by coarse spatial resolution and imperfect aerosol and cloud parameterization; post-processing methods such as MOS or statistical downscaling can substantially reduce forecast errors particularly when correcting aerosol-related biases and improving terrain representation.

4.2. Statistical methodology

Statistical approaches encompass a number of traditional statistical methodologies, including regression analysis [64], autoregressive models [65], Markov Chain [66], and machine learning methodologies, such as support vector machine (SVM) [67] and ANN [68]. Accuracy of the statistical approaches is highly contingent upon the historical data and bears no relation to the internal state information of the solar system [69]. By evaluating all accuracy metrics, including RMSE, MBE, MAE, and the coefficient of determination (R^2), it is evident that all machine learning models enhance the accuracy of regional GHI predictions [70]. Comparative analyses of various statistical models including Neural Networks (NN), Multivariate Regression (MVR), and Support Vector Machines (SVM) using Mean Absolute Percentage Error (MAPE) and Mean Squared Error (MSE) criteria for short-term (one-hour-ahead) irradiance forecasting based on ground-measured meteorological inputs (temperature, humidity, and cloud index) indicate that NN models slightly outperform MVR and SVM under similar data conditions [71]. However, this superiority depends on the forecast horizon, data quality, and local climatic variability.

To investigate the amount of incoming radiation in various countries under the influence of different parameters, numerous studies have employed this method. For instance, Sabzi Peror et al. [72] conducted a study to examine various influential parameters on the monthly average solar radiation index, considering factors such as altitude, the amount of suspended dust on different days of the month, and the seasonal variation in the Earth-Sun distance. They developed an altitude-dependent method incorporating these parameters and tested it in the central arid deserts of Iran. The results demonstrated that this modified altitude method achieved an error rate of $<2\%$ and exhibited high prediction accuracy in arid and semi-arid deserts globally, with effective generalization to the conditions in Iran.

Olimar et al. [73] conducted an investigation into the changes in

solar radiation in the southern region of Algeria, taking into account the climatic conditions. They analyzed 10 years of solar radiation data from 2011 to 2020. Their findings revealed significant seasonal and annual variability, highlighting the region's substantial solar energy potential, with an average daily energy measurement exceeding 6.16 kWh/m^2 and an annual average of 2.2 MWh/m^2 .

In highly accurate solar power forecasting using statistical methods, it is important to reduce the risk of serious errors (especially over-estimation). Takamatsu et al. [74] addressed this subject by developing low risk models with good average accuracy through the following approach: adoption of a machine learning ensemble, and optimization of the metamodel using quadratic programming with downsampled validation data.

4.2.1. Regression analysis

Regression analysis is an approach in forecasting solar irradiance. For instance, a linear regression model is introduced as a post-processing method to minimize errors from the Weather Research and Forecasting (WRF) model. The coefficients of the regression model are updated using the Kalman Filter (KF) approach, enabling the model to adjust dynamically to changes in solar radiation. The KF updating formula is then altered to account for measurement availability constraints during forecast execution. WRF forecasts include inputs such as radiation, temperature, relative humidity, and solar zenith angle. Fig. 7 shows a logistic distribution censored at zero [75].

In another work, Lauret et al. [76] developed simple linear models that allow users to predict solar irradiance forecast errors based solely on solar variability at a specific location on Earth. Sixty sites distributed around the globe were used to build two types of RMSE prediction models. The first type was for intra-day forecasts (1-hour to 6-hour forecast horizons), while the second is for day-ahead forecasts (24-hour horizon). The derivation of the intra-day forecast error prediction model leverages on a non-linear time series approach whereas the one for day-ahead forecast error relies on forecasts issued by the European Centre for Medium-Range Weather Forecasts (ECMWF).

Khan et al. [77] introduced novel hybrid architectures and presented a detailed evaluation of GHI prediction models. The approach included traditional machine learning algorithms (Support Vector Regressor, Decision Tree Regressor, and Random Forest Regressor) and novel hybrid deep learning architectures (BiLSTM-BiGRU and CNN-LSTM). The novel cascaded BiLSTM-BiGRU architecture used complete bidirectional processing. The CNN-LSTM framework effectively combined spatial and temporal features. These advanced architectures achieved a 30 % reduction in computational complexity compared to equivalent standalone deep learning approaches, enhanced robustness to missing or noisy data, and controlled seasonal variations.

4.2.2. Artificial intelligence techniques

Throughout the history of solar energy forecasting, ambient satellites have measured solar energy reflected from the Earth or atmosphere. Consequently, satellite-based methods are now well-established, and extensive research has been conducted to address their limitations [78]. Recent studies have also incorporated machine learning methods and artificial intelligence techniques.

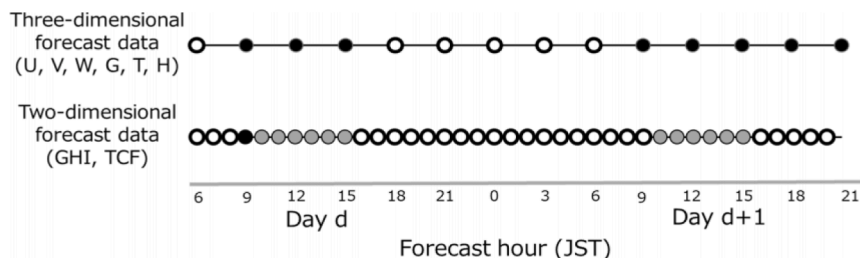


Fig. 6. CI of the NWP forecast data for days (d) and ($d + 1$) [64].

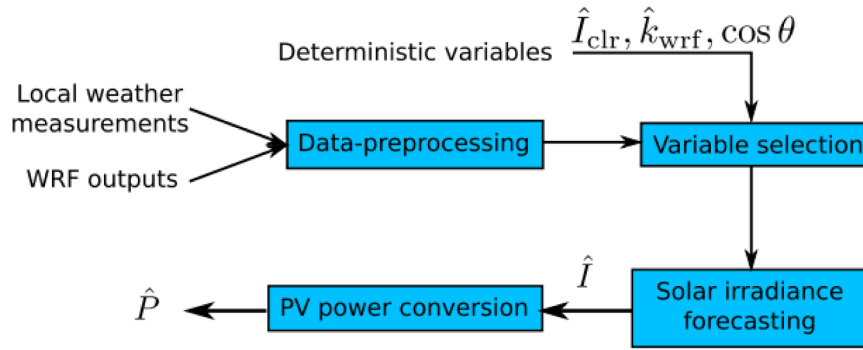


Fig. 7. Proposed scheme of one-day-ahead PV power forecasting [75].

To demonstrate the positive impact of employing artificial intelligence techniques, Khadija Barhmi et al. [79] conducted a comprehensive review of various solar irradiance forecasting methods based on satellite data, all-sky imagery (ASI), sensor networks, NWP models, and hybrid approaches. By analyzing more than eighty studies, the authors proposed a comprehensive framework for the classification and evaluation of forecasting methods, utilizing common performance indicators such as RMSE, MAE, MBE, and the forecast skill index (FS) to assess model accuracy. Their findings reveal that the integration of artificial intelligence particularly deep neural networks (CNN, LSTM, GAN) and hybrid learning strategies significantly enhances forecasting precision over short- to medium-term horizons. Furthermore, the study underscores the crucial role of data preprocessing for noise reduction and postprocessing for bias correction and performance refinement, highlighting how AI-driven methodologies can profoundly elevate the accuracy and reliability of solar forecasting systems.

To evaluate the effectiveness of artificial intelligence-based approaches, numerous studies have compared their performance against traditional forecasting methods under identical operating conditions. For instance, Shanlin Chen et al. [80] investigated a deep learning-based model chain developed for intra-day solar irradiance forecasting. In the first stage, spectral satellite data from GOES-16 and

ground-based measurements from the SURFRAD network were used to train a model capable of producing highly accurate spatial estimations of GHI, referred to as the SAT-DL model. In the subsequent stage, these spatial estimates served as inputs to a second model designed to generate short-term temporal forecasts of GHI (ranging from 15 to 180 min ahead) using convolutional neural networks (CNNs), attention layers, and dense layers within the TensorFlow framework. For performance benchmarking, NSRDB data derived from physical radiative transfer models were employed. Evaluation metrics such as nRMSE, nMBE, the coefficient of determination (R^2), and ramp event analysis demonstrated that the proposed deep learning model chain outperformed benchmark models (including hybrid physical-deep learning and end-to-end networks), particularly under cloudy conditions. Although the approach demands greater computational resources and access to ground-based data, it offers substantial potential for enhancing solar forecasting accuracy and advancing intelligent energy management systems.

In this context, Sediqi et al. [81] utilized time-lapse videos from a wide-lens camera, analyzing one million images spanning 1 to 12 years, to predict solar radiation using a deep learning approach. Fig. 8 illustrates the Convolutional Neural Network and Long Short-Term Memory (CNN-LSTM) model proposed by these researchers. This method comprises two stages: the first stage involves a CNN to encode a frame from a

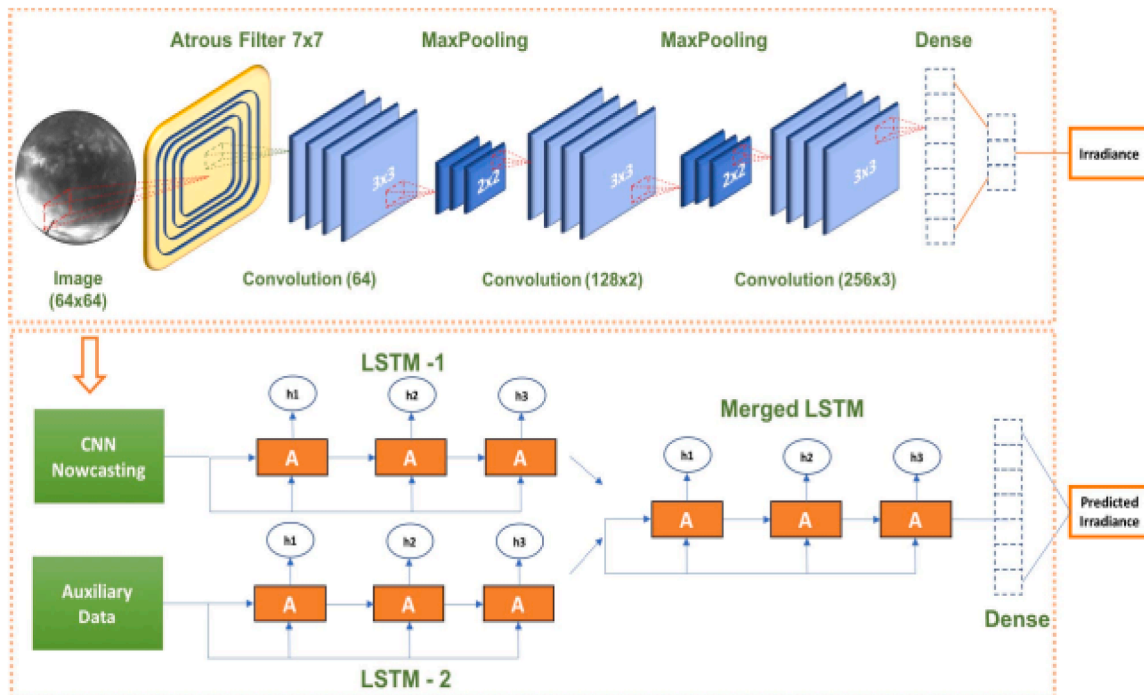


Fig. 8. The proposed CNN + LSTM approach based architecture [77].

sky video, creating a full-sky representation with the help of auxiliary weather data. The second stage employs a two-tier LSTM architecture to analyze historical full-sky representations and generate future predictions. This deep learning approach, when compared to satellite-based methods, significantly lowers the normalized mean absolute percentage error for both nowcasting (predicting solar irradiance at the moment the frame is captured) and forecasting (predicting irradiance up to 4 h in advance).

Fathi et al. [82] also assessed the viability of a rooftop solar photovoltaic system for a hotel building and utilized monthly data from 1984 to 2023 and hybrid models comprising LSTM-based and GRU-based architectures, along with the MLP model and the standard forecasting methods, to forecast solar radiation and temperatures.

Forecast accuracy directly impacts operational efficiency each 1 % reduction in forecasting error (RMSE) can translate into approximately 0.5–1 % improvement in PV plant energy scheduling efficiency and reduced reserve capacity costs, leading to more stable grid integration. Xiaoyan Shao et al. [83]. shown that more accuracy can be achieved by combining the individual forecasting models using a machine learning model. In Fig. 9, schematic of this proposed combining is represented.

However, existing statistical and machine learning algorithms are not reliable for forecasting due to the sparse nature of solar energy data. Several factors affect the performance of solar radiation, such as the forecast horizon, weather classification, and performance evaluation criteria. Therefore, an overview of the performance of deep learning-based solar radiation forecasting models is presented in separate classifications. Based on the analysis, deep learning models outperform conventional models in solar forecasting applications, especially in combination with some techniques that enhance feature extraction. In addition, the use of data augmentation techniques is beneficial to improve the performance of deep learning, especially for deep networks [84].

4.2.3. Horizon-based suitability

Recent comparative studies show that image-based deep learning (CNN, ConvLSTM) and optical-flow methods excel at very-short term (minutes to 1 hour) nowcasting, while NWP-augmented ML or hybrid physical-ML methods perform better for horizons >1–6 h and up to multi-day forecasts. For example, Ajith et al. [85] report superior performance of sky-image DL for sub-hour horizons, whereas satellite+NWP+DL pipelines reduce error for intra-day and day-ahead

horizons.

To address the high levels of uncertainty inherent in photovoltaic energy, an increasing number of studies have focused on short-term or nowcasting solar forecasting. Most of these works employ deep learning models to directly predict solar irradiance or photovoltaic power based on sequences of sky images. Recently, however, advances in generative modeling have introduced a new paradigm that divides the nowcasting problem into two sub-tasks: (1) future event prediction, involving the generation of forthcoming sky images; and (2) irradiance or power estimation, which predicts solar values from a single image.

Among these approaches, the SkyGPT model has shown that its strength lies primarily in the estimation component rather than in image generation. Building on this, Lorenzo F. C. Varaschin et al. [86] concentrated on the irradiance estimation task, systematically evaluating a wide range of deep learning architectures using the well-known Folsom, SIRTa, and NREL datasets. Their work also included ablation studies to test different training setups and data preprocessing techniques, such as the choice of target variables and time-label alignment between sky images and irradiance measurements. Notably, they identified a potential timestamp misalignment in the Folsom dataset and proposed a correction strategy. Across all three datasets, their results consistently confirmed the reliability and generalizability of their findings across different solar observation sites.

In summary, it can be stated:

- (1) Very-short term (seconds–60 min): sky-image DL and optical-flow-based methods yield the best nowcasting accuracy.
- (2) Short term (1–6 h): hybrid ML+NWP and post-processed NWP (AnEn, MOS) provide better forecasts by leveraging synoptic evolution.
- (3) Medium to long term (>6 h-days): global NWP ensembles (ECMWF) with statistical downscaling remain the primary choice. These distinctions are supported by comparative studies and benchmarks [85,86]

4.2.4. Climate/Regional suitability

Model performance is strongly influenced by climatic conditions. In particular, NWP models tend to show varying accuracy depending on the dominant regional weather regimes. For example, coastal and stratocumulus-dominated regions often pose significant challenges for conventional cloud microphysics and cloud-radiation parameterizations within NWP systems. In such areas, models can benefit substantially from higher-resolution simulations (e.g., WRF) combined with advanced statistical or machine-learning-based postprocessing techniques to better represent local cloud variability and short-term solar irradiance fluctuations [51].

Conversely, in arid and semi-arid regions, where dust storms and other aerosol events frequently occur, the dominant source of uncertainty shifts from cloud representation to aerosol–radiation interactions. In these environments, improved skill is obtained through aerosol-aware modeling approaches or the use of aerosol data assimilation methods that integrate satellite or ground-based observations [80].

Finally, machine learning (ML) models that are trained on local sky imagery or radiation patterns often exhibit limited transferability across different climatic zones. This is because the spectral, radiative, and cloud-formation characteristics can vary dramatically between humid coastal areas and dry continental regions. Therefore, to maintain generalization and model robustness, regional retraining, climate-specific adaptation, or transfer-learning techniques are strongly recommended [87].

4.2.5. Complexity vs latency tradeoffs

There exists a practical trade-off between model complexity and operational latency in predictive modeling. More sophisticated architectures such as Transformers, ConvLSTMs, or multimodal ensemble models often achieve lower RMSE (higher predictive accuracy).

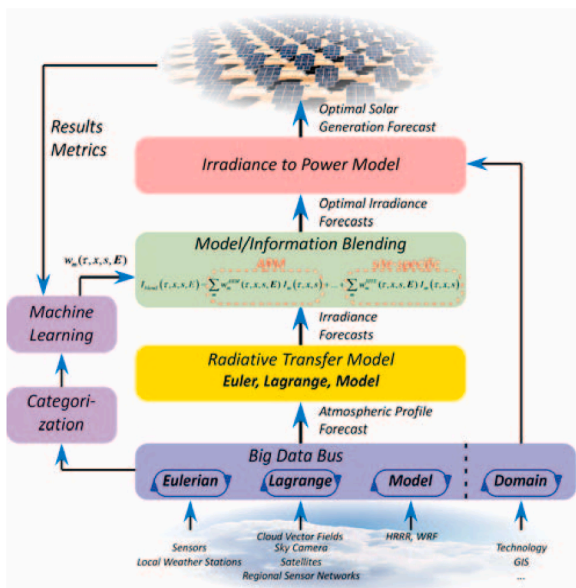


Fig. 9. General schematic showing the architecture of the machine learning based model blending forecast system [78].

However, these gains usually come at the cost of greater computational requirements and higher inference latency, which can limit their usability in real-time or resource-constrained environments.

In contrast, simpler approaches including persistence models, ARIMA, or lightweight Random Forests may deliver less accurate predictions, but they are typically faster, more stable, and easier to deploy in operational systems where low latency or limited computational capacity is critical [86,88].

4.2.6. Neural network

In forecasting solar radiation, ANN are utilized because they can learn from historical data to make the required predictions. The key input data influencing the availability and intensity of solar radiation include the month, day, season, average ambient temperature, and average relative humidity (RH) [89]. A generalized regression neural network is used to predict solar radiation. For training the neural network, meteorological satellite and geographical data were employed for short-term and long-term periods. The network's input layer were meteorological satellite and geographic data such as longitude, latitude, altitude, month, and average earth surface temperature, with solar radiation as the output. The root mean square error and the correlation coefficient between the predicted and actual values were calculated using the neural network outputs [90].

Various models have been created and evaluated to forecast hourly solar radiation one day ahead using artificial neural network techniques. The projected radiation is the primary factor affecting the panel array of modules in a photovoltaic power plant. When comparing pure statistical models, which depend solely on locally measured data, with methods that adjust numerical weather prediction data using a stability model, it is found that statistical models can improve forecasting performance [91,92].

Cloud cover is one of the most critical variables in solar irradiance forecasting. Accurate predictions of cloud cover changes can provide valuable information for companies and systems utilizing solar energy. McCandless et al. [93] proposed a short-range solar irradiance forecasting method that uses a statistical classification of cloud regimes to enhance deterministic GHI predictions. Their study utilized surface weather observations, solar irradiance measurements, and GOES-East satellite data. The prediction method utilized a regime-dependent approach to forecast over intervals of 0, 60, 120, and 180 min. To

calculate the clearness index, various inputs and multi-regime dependent prediction methods were employed. Fig. 10 illustrates the relationship between these techniques. From left to right, the first prediction method is the ANN, applied to the clear dataset. The second is an ANN without additional regime classification. The remaining methods are RD-ANNs. The first RD-ANN method, called RD-ANN-GCT, is based on regimes identified by cloud type changes in the GOES-East data. The next method, RD-ANN-GKTCC, uses k-means regimes incorporating inputs from all data sources in this study. The final technique, named RD-ANN-KtCC, involves satellite measurements and compares previous work.

Zibo Dong et al. [94]. investigated the hourly solar irradiance time series using satellite image analysis combined with a hybrid exponential smoothing state space (ESSS) model and an ANN. The ESSS model was employed to predict the cloud cover index, given the stochastic nature of cloud formation. To estimate solar irradiance from the cloud index, a MLP was selected from various ANN models. Satellite images were sourced from the Geostationary weather satellite operated by the Japan Meteorological Agency (JMA), and the solar irradiance time series data were obtained from the meteorological station of the Solar Energy Research Institute of Singapore (SERIS). The satellite images collected were thoroughly analyzed to determine the cloud cover index using self-organizing maps. Two critical threshold tests were then applied to the satellite images: the visible reflectance threshold test and the infrared temperature threshold test. For the visible reflectance threshold test, image data from one year were utilized to create monthly synthetic raster images of surface albedo. The infrared temperature threshold test used two temperature sources: the monthly minimum sea surface temperature and the monthly mean minimum land temperature from the National Environment Agency of Singapore, to generate monthly synthetic raster images of surface temperature. Fig. 11 displays the normalized RMSE for all models. The proposed model in this study demonstrated a 6 % improvement in accuracy compared to other models.

4.3. Combined method

Ensemble models, due to their reduced constraints compared to individual models, often yield more accurate results than statistical and physical models. Over the past three decades, numerous researchers

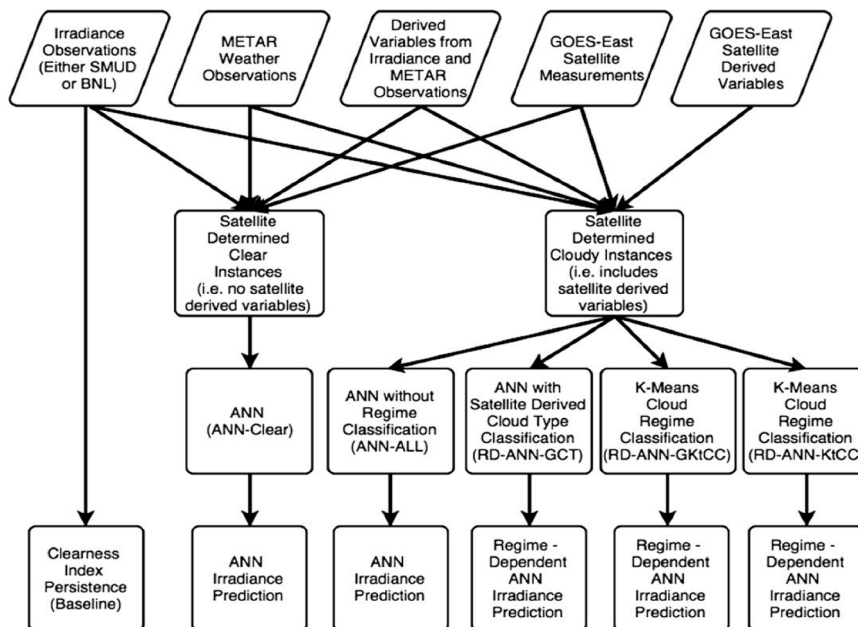


Fig. 10. Overall process design for our regime-dependent prediction technique and the comparison techniques [83].

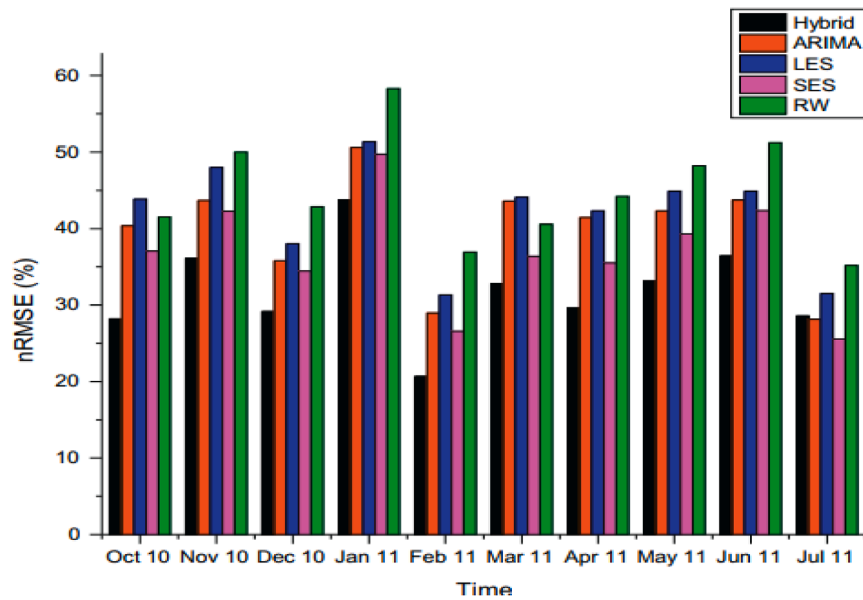


Fig. 11. nRMSE (%) comparison among 5 forecasting methods for SERIS data [84].

have continuously strived to enhance the accuracy of solar radiation predictions. This section outlines methods for estimating solar energy potential at specific locations by integrating various solar radiation calculation techniques. This innovative approach, which incorporates regional climatic conditions, significantly reduces computation time and enhances decision-making accuracy with minimal initial data. Additionally, it examines the impact of spatial orientation on the maximum electricity generation over monthly, seasonal, and annual periods, demonstrating that optimal tilt angles during specific times can maximize the region's solar potential [95]. As an example, surface meteorological data from the DATSAV2 database allow for the application of the METSTAT (meteorological/statistical) model to compute hourly values of normalized direct, horizontally diffuse, and global horizontal solar radiation for locations worldwide [96].

To forecast GHI, researchers have proposed several hybrid methods. Marquez et al. [97] utilized processed satellite images in combination with an ANN to develop a new hybrid approach for predicting GHI at intervals of 30, 60, 90, and 120 min. The forecast model was applied to GHI data collected from Davis and Merced locations in the San Joaquin Valley. The ANN model inputs included cloud fraction variables derived from satellite image analysis, such as velocimetry and cloud indexing. The proposed forecasting methods showed a relative improvement ranging from 5 % to 25 %. Additionally, as the forecasting horizon increased, the RMSE improvement also increased. In another study, de Lima et al. [98] predicted GHI over short- and medium-term horizons using the Global Forecast System (GFS), the Weather Research and Forecasting-Solar (WRF-Solar) model, and a hybrid method called GFS-MOS for the Northeast of Brazil. The GFS-MOS combines GFS forecasts with the Model Output Statistics (MOS) algorithm, which uses observational data for statistical refinement. The results demonstrate that GFS-MOS outperforms WRF-Solar in most aspects, particularly in simulating low GHI values, where GFS-MOS demonstrated superior accuracy.

Post-processing methods have widely been employed as correction models to enhance the accuracy of NWP solar irradiance forecasts. Based on the analysis of NWP GHI error characteristics, errors in NWP GHI forecasts may show obvious seasonal variations. Therefore, it is challenging for a single correction model to achieve good correction performance and strong seasonal robustness. Hence, Dou et al. [99] proposed a hybrid model comprising representation learning module, feature sparse activation module, and encoder-decoder-based correction

module to address the aforementioned problem.

Ansong et al. [100] presented a hybrid deep learning model combining convolutional neural networks (CNN) with long short-term memory (LSTM) networks to forecast solar irradiance 60 min ahead. The model used images from the Karlsruhe low-cost sky imager (KALISI) alongside meteorological data from a local weather station. Experimental results showed that including relative humidity and wind direction yields the lowest average RMSE of 119 W/m² and MAE of 82 W/m². This outperformed image-only and persistence models by 3 % and 42 % in RMSE (9 % and 50 % in MAE), respectively.

A new combined Model Output Statistics (MOS) method, known as the MOS cascade, was created to improve the global horizontal radiation forecasts produced by the WRF model. This approach used a sequence of two different MOS algorithms. The first algorithm, MOSRH, is a physically-based method aimed at correcting moisture-related errors in WRF radiation schemes. The second algorithm, MOSNN, leverages artificial intelligence to address systematic and learnable errors in numerical weather predictions. By comparing 1-day and 2-day forecast accuracy with observed radiation data, a significant reduction in error was observed when using the MOSRH model and the MOS cascade [101].

Dong et al. [102] proposed a novel zero-shot multimodal learning framework for predicting global horizontal irradiance. The proposed framework integrates satellite images and numerical time series data. First, a pre-trained Vision Transformer was employed to extract feature embeddings from satellite images. Meanwhile, the temporal information from historical irradiance and relevant variables are captured using an iTransformer. Second, a filtering cross-attention mechanism was introduced to eliminate inter-modal redundant information and enhance the modal fusion capability. Finally, a multivariate shared multi-layer perceptron (MLP) was used to achieve multi-step prediction. This paper introduces a method based on fully convolutional neural networks (U-Net) to interpolate missing time steps for both deterministic and probabilistic solar irradiance forecasts. The model is trained on deterministic forecasts (ECMWF IFS) to restore hourly forecasts from three-hourly forecasts benefiting from past physical NWP model runs. We show that the proposed method is capable of restoring deterministic forecasts and furthermore can provide highly accurate probabilistic solar irradiation forecast interpolations.

Real-time solar radiation predictions can be made using either machine learning or physics-based models, each offering unique benefits

and drawbacks. To overcome these limitations, a new predictive analytics framework was introduced, combining the strengths of both model types. This framework, particularly a thrust conceptual model, integrates satellite cloud imagery with real-time solar radiation data to improve prediction accuracy. Initial results, derived from actual radiation and cloud data, validate the effectiveness of this conceptual drift-based approach. The findings suggest that static models are inadequate for accurately capturing solar radiation dynamics under both cloudy and clear conditions [103].

Solar energy is extensively used in both residential and industrial applications. To maximize its utilization, measuring radiation per unit area is crucial. Irradiance measurements are taken at numerous stations. When radiation data is missing, empirical models are often used. For instance, Holland et al. [104] created a 48-hour forecasting system for two PV power plants in China and Germany. This system combines NWP models with irradiance and power measurements, along with satellite-based irradiance forecasts. Various GHI forecasts were merged into a single forecast using linear regression, which was then applied to PV simulation. The resulting PV power predictions were compared with on-site power measurements, and Root Mean Square Error was used to assess the accuracy of the irradiance and PV power predictions (Fig. 12).

The conventional method for photovoltaic forecasting depends heavily on extensive historical data, including records of PV plant power generation and numerical weather prediction data, which makes it unsuitable for newly established PV plants. To overcome this limitation, a physical forecasting approach based on solar radiation on inclined surfaces was proposed, as depicted in Fig. 13. This method employs three decomposition models and four displacement models, resulting in a total of 12 combined forecasting models. The enhanced model also took into account factors such as solar spectral response, radiation angle, and soil coefficient. The predicted PV power was then calculated using these derived prediction models along with the mathematical model of the PV cell. The performance of the 12 models was assessed using RMSE and

MAE under four different weather conditions: sunny, cloudy, overcast, and rainy days. The evaluation revealed that the Liu-Jordan transfer model combination methods demonstrated superior prediction accuracy across these weather conditions. Notably, the Erbs + Liu-Jordan model yielded the most accurate predictions [105].

Machine learning techniques have become increasingly popular in various forecasting applications. For example, Uğur Ercan [106] focused on predicting solar radiation for all cities in Turkey's Mediterranean region using both statistical and common machine learning methods. The performance of these models was evaluated using ensemble learning methods such as gradient boosting and extreme gradient boosting, along with artificial neural networks. Data from eight cities in the Mediterranean region of Turkey were utilized, with input variables including deviation angle, wind speed, ambient temperature, relative humidity, and cloud index. The findings reveal that machine learning models outperformed statistical models. The extreme gradient boosting method delivered the best results, achieving an R^2 of 0.9993 and a MAPE of 0.0119.

4.4. Pros and cons of physical vs AI models and the importance of hybrid approaches

Physical models (NWP) are physically consistent and suited to medium-to-long horizons but suffer from cloud/aerosol parameterization and coarse resolution; AI models (ML/DL) excel at capturing local nonlinearities and very short-term variability but require representative training data and may not generalize across climates. Hybrid methods that fuse NWP/satellite physics with ML postprocessing or multimodal DL provide a pragmatic compromise improving accuracy and reducing systematic biases, as demonstrated in recent studies [88].

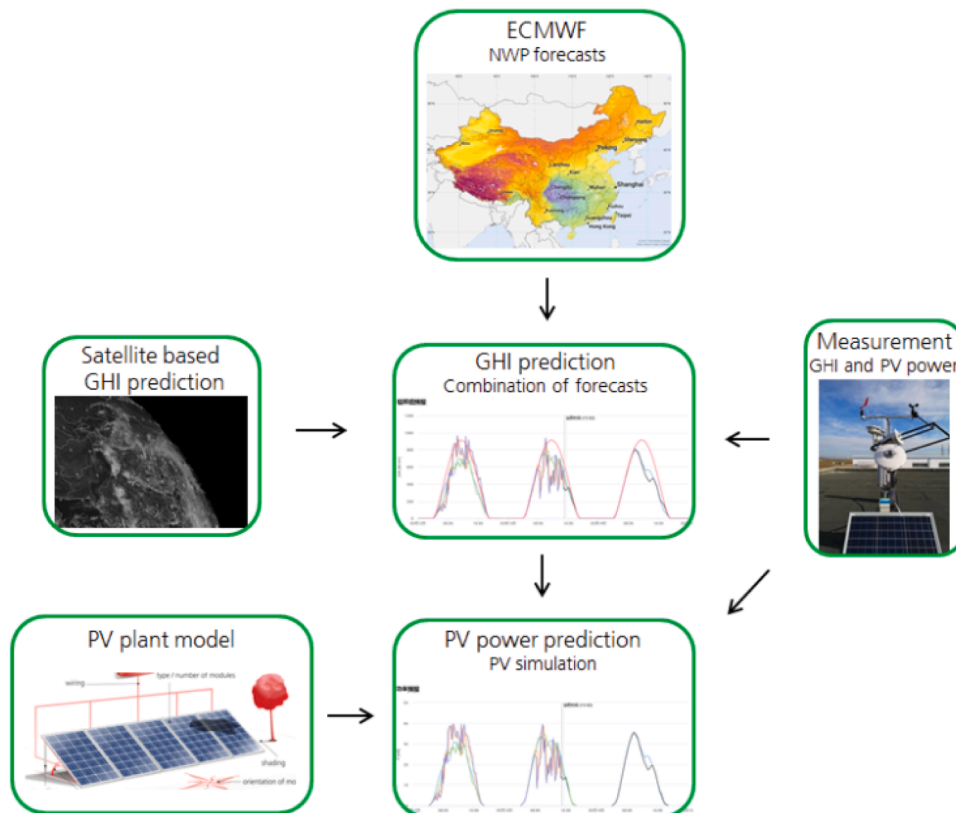


Fig. 12. Integration of multiple GHI predictions as simulation input and comparison with measured power [104].

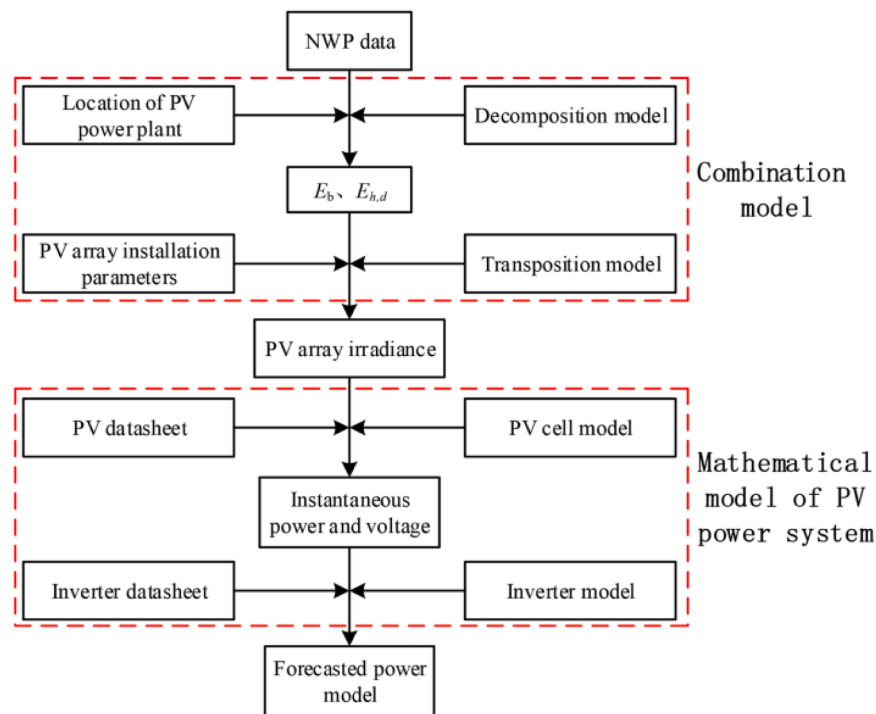


Fig. 13. Structure diagram of the PV power forecasting [91].

5. Conclusion

Solar irradiance forecasting is pivotal for the integration of solar energy into the power grid, as it addresses the inherent variability and uncertainty associated with solar power generation. This review has examined the progression of forecasting methods from traditional physical models to modern data-driven approaches, highlighting the strengths and limitations of each technique.

Physical models, such as Numerical Weather Prediction, offer a foundational understanding of atmospheric processes but often fall short in accuracy for short-term forecasts due to their reliance on large-scale atmospheric data. Statistical models, while useful in identifying patterns in historical data, tend to struggle with capturing the nonlinear and dynamic nature of solar irradiance, especially over short time horizons. In contrast, AI and ML methods, particularly deep learning, have shown great promise in improving forecast accuracy by effectively handling complex, nonlinear relationships between meteorological variables.

The review also emphasized the importance of integrating various data sources, such as satellite imagery and ASIs, into forecasting models. These integrations help to overcome the limitations of individual data sources and enhance the overall prediction accuracy, particularly for short-term and very short-term forecasts. Hybrid approaches that combine physical models with AI techniques have emerged as particularly effective, leveraging the strengths of both to produce more reliable forecasts.

Despite these advancements, challenges remain, particularly in the accurate forecasting of solar irradiance under varying atmospheric conditions, such as cloud cover and aerosol presence. The review underscores the need for continued research into improving the resolution and accuracy of forecasting models, including the development of more sophisticated hybrid models and the integration of real-time data for adaptive forecasting.

In conclusion, while significant progress has been made in solar irradiance forecasting, ongoing advancements in AI, data integration, and hybrid modeling are essential to further enhance the reliability and precision of these forecasts. Future research should focus on refining these methods, exploring new data sources, and addressing the

challenges posed by complex atmospheric phenomena to support the growing reliance on solar energy in global power grids.

CRediT authorship contribution statement

Mohammad Hasan Ghoduseinejad: Writing – review & editing, Project administration, Methodology, Investigation, Conceptualization. **Nasrin Rashvand:** Writing – original draft, Visualization, Software, Investigation. **Fatemeh Salmanpour:** Writing – original draft, Methodology, Investigation, Conceptualization. **Shaghayegh Danehkar:** Writing – original draft, Methodology, Investigation, Conceptualization. **Hossein Yousefi:** Writing – review & editing, Supervision, Project administration, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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